

# How to create a report for multi-select variables in 4 steps

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## ABSTRACT

Using the power of the DATA Step, SAS® Macro Language, and the SQL Procedure, I was able write efficient, parsimonious code to summarize the frequency of use of twenty-two drugs at four discrete time periods relative to a diagnosis date.

## PRESENTATION OF CHALLENGE

Recently, while assigned to a project in which I was the lead programmer, I was tasked with creating a report, from a total of 88 variables, summarizing the use of 22 drugs taken by patients within any of 4 time periods relative to a diagnosis date (DX\_DT): 12 months prior to, 0 to 3 months, 3 to 6 months, and/or 6 to 12 months following diagnosis. Each of the 88 variables represented a flag for use of a particular drug, within one of the time periods, where the value of the variable could either be 1 (if a drug was taken within a given time period) or 0 (if a drug was not taken within a given time period). The denominator for the table was fixed.

Figure 1. Table summarizing frequency of drug use.

NESUG 2013

Table of drugs  
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| NESUG 2013: Frequency of use of medications prior to and following diagnosis date |                                         |                                              |                                                                  |                                                                  |
|-----------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|
| Frequencies of use of each medication<br>Within the following time periods:       | 12 months prior to<br>diagnosis<br>N=44 | 0 to 3 months<br>following diagnosis<br>N=44 | More than 3<br>months to 6 months<br>following diagnosis<br>N=44 | More than 6 months<br>to one year<br>following diagnosis<br>N=44 |
| Drug A<br>Brand Name A                                                            | 10 (22.73%)                             | 15 (34.09%)                                  | 12 (27.27%)                                                      | 13 (29.55%)                                                      |
| Drug B<br>Brand Name B                                                            | 3 ( 6.82%)                              | 5 (11.36%)                                   | 4 ( 9.09%)                                                       | 3 ( 6.82%)                                                       |
| Drug C<br>Brand Name C                                                            | 15 (34.09%)                             | 17 (38.64%)                                  | 18 (40.91%)                                                      | 20 (45.45%)                                                      |
| Drug D<br>Brand Name D                                                            | 2 ( 4.55%)                              | 1 ( 2.27%)                                   | 2 ( 4.55%)                                                       | 3 ( 6.82%)                                                       |
| Drug E<br>Brand Name E                                                            | 13 (29.55%)                             | 4 ( 9.09%)                                   | 4 ( 9.09%)                                                       | 5 (11.36%)                                                       |
| Drug F<br>Brand Name F                                                            | 0 ( 0.00%)                              | 0 ( 0.00%)                                   | 0 ( 0.00%)                                                       | 0 ( 0.00%)                                                       |
| Drug G<br>Brand Name G                                                            | 12 (27.27%)                             | 8 (18.18%)                                   | 6 (13.64%)                                                       | 8 (18.18%)                                                       |
| Drug H<br>Brand Name H                                                            | 1 ( 2.27%)                              | 1 ( 2.27%)                                   | 1 ( 2.27%)                                                       | 2 ( 4.55%)                                                       |
| Drug I<br>Brand Name I                                                            | 4 ( 9.09%)                              | 2 ( 4.55%)                                   | 1 ( 2.27%)                                                       | 1 ( 2.27%)                                                       |
| Drug J<br>Brand Name J                                                            | 4 ( 9.09%)                              | 3 ( 6.82%)                                   | 2 ( 4.55%)                                                       | 1 ( 2.27%)                                                       |

Figure 2. Snapshot of data used to create table displayed in Figure 1.

|    | PATIENT_ID | FLAG | DX_DT     | DRUGA1YRPREV | DRUGA3 | DRUGA3TO6 | DRUGA6TO12 | DRUGB1YRPREV | DRUGB3 | DRUGB3TO6 | DRUGB6TO12 |
|----|------------|------|-----------|--------------|--------|-----------|------------|--------------|--------|-----------|------------|
| 1  | 1001300    | 1    | 11OCT2011 | 0            | 1      | 1         | 1          | 0            | 0      | 0         | 0          |
| 2  | 1001518    | 1    | 09FEB2009 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 3  | 1001614    | 1    | 30NOV2010 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 4  | 1002099    | 1    | 08JUL2010 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 5  | 1003210    | 1    | 04NOV2009 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 6  | 1003503    | 1    | 07MAY2009 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 7  | 1003537    | 1    | 02DEC2009 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 8  | 1004165    | 1    | 21JUL2008 | 1            | 1      | 1         | 1          | 0            | 0      | 0         | 0          |
| 9  | 1004666    | 1    | 18JUN2009 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 10 | 1005277    | 1    | 08APR2010 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 11 | 1005962    | 1    | 01JUL2010 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 12 | 1006237    | 1    | 24SEP2008 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 13 | 1006902    | 1    | 24MAR2011 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 14 | 1007284    | 1    | 02MAR2010 | 0            | 1      | 1         | 1          | 0            | 0      | 0         | 0          |
| 15 | 1007720    | 1    | 08JUL2010 | 0            | 0      | 0         | 1          | 0            | 0      | 0         | 0          |
| 16 | 1007779    | 1    | 16JUN2010 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 17 | 1008207    | 1    | 13MAR2011 | 0            | 0      | 1         | 1          | 0            | 0      | 0         | 0          |
| 18 | 1008498    | 1    | 18JUL2010 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 19 | 1008586    | 1    | 12AUG2008 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 20 | 1009655    | 1    | 07MAY2010 | 1            | 1      | 0         | 0          | 0            | 0      | 0         | 0          |
| 21 | 1009874    | 1    | 16SEP2010 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 22 | 1010064    | 1    | 16MAY2009 | 1            | 1      | 0         | 0          | 0            | 1      | 1         | 1          |
| 23 | 1010737    | 1    | 17NOV2009 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 24 | 1011741    | 1    | 04FEB2010 | 0            | 1      | 1         | 1          | 0            | 0      | 0         | 0          |
| 25 | 1012917    | 1    | 12APR2010 | 0            | 0      | 0         | 0          | 1            | 1      | 1         | 1          |
| 26 | 1013675    | 1    | 01OCT2010 | 1            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 27 | 1013885    | 1    | 27APR2010 | 1            | 1      | 0         | 0          | 0            | 0      | 0         | 0          |
| 28 | 1015145    | 1    | 29OCT2010 | 0            | 0      | 0         | 0          | 1            | 1      | 1         | 0          |
| 29 | 1016966    | 1    | 21SEP2009 | 0            | 0      | 0         | 0          | 0            | 1      | 1         | 1          |
| 30 | 1017154    | 1    | 14APR2010 | 0            | 0      | 0         | 0          | 0            | 0      | 0         | 0          |
| 31 | 1017297    | 1    | 22DEC2009 | 1            | 1      | 0         | 0          | 0            | 0      | 0         | 0          |

Using Drug A as an example, four flags existed, identifying use of Drug A at 12 months prior to, up to 3 months after, 3 to 6 months after, and 6 to 12 months after the diagnosis date.

| <u>Drug</u> | <u>Flag</u>                                              |
|-------------|----------------------------------------------------------|
| Drug A      | DRUGA1YRPREV (12 months prior to the diagnosis date)     |
|             | DRUGA3 (up to 3 months following the diagnosis date)     |
|             | DRUGA3to6 (3 to 6 months following the diagnosis date)   |
|             | DRUGA6to12 (6 to 12 months following the diagnosis date) |

The goal was to summarize the frequency of use for each drug, within each time period. Additionally, based on the table shell provided for the report, a summary of drug usage needed to be displayed in alphabetical order. One could compute counts and percentages using the FREQ Procedure, and wrap the code used to generate these summaries in a macro; however, the macro would have to be invoked 88 times, once for each variable. Since I wanted to write the most parsimonious, yet effective, code possible, I decided to harness the power of the SAS® DATA Step, SAS® Macro Language, and the SQL Procedure to create a relatively simple, yet effective, solution.

## PROPOSED SOLUTION

First, four temporary datasets were created, one for each time period. For example, all variables corresponding to flags for drug use within the 1<sup>st</sup> time period were in the dataset labeled “COL1\_VARS”, all variables with flags corresponding to drug use within the 2<sup>nd</sup> time period were in the dataset labeled “COL2\_VARS”, and so on, until four datasets were created. To ensure that the variables were ordered alphabetically, I used the ATTRIB statement within the DATA Step.

#### ATTRIB

```
DRUGA1YPREV FORMAT = BEST12.  
DRUGB1YPREV FORMAT = BEST12.  
DRUGC1YPREV FORMAT = BEST12.  
DRUGD1YPREV FORMAT = BEST12.  
DRUGE1YPREV FORMAT = BEST12.  
DRUGF1YPREV FORMAT = BEST12.  
DRUGG1YPREV FORMAT = BEST12.  
DRUGH1YPREV FORMAT = BEST12.  
DRUGI1YPREV FORMAT = BEST12.  
DRUGJ1YPREV FORMAT = BEST12.  
DRUGK1YPREV FORMAT = BEST12.  
DRUGL1YPREV FORMAT = BEST12.  
DRUGM1YPREV FORMAT = BEST12.  
DRUGN1YPREV FORMAT = BEST12.  
DRUGO1YPREV FORMAT = BEST12.  
DRUGP1YPREV FORMAT = BEST12.  
DRUGQ1YPREV FORMAT = BEST12.  
DRUGR1YPREV FORMAT = BEST12.  
DRUGS1YPREV FORMAT = BEST12.  
DRUGT1YPREV FORMAT = BEST12.  
DRUGU1YPREV FORMAT = BEST12.  
DRUGV1YPREV FORMAT = BEST12.  
;
```

Briefly, the ATTRIB statement is an assignment statement that can be used to permanently assign lengths, formats, and informats to pre-existing variables by changing descriptor information in a dataset. If the variables don't exist, the ATTRIB statement, in conjunction with subsequent assignment statements, can be used for to create variables.

Once the four temporary datasets were created, I stored the names of each of the variables into macro variables. There are a number of ways macro variables can be created, such as using the %LET statement, or CALL SYMPUT/CALL SYMPUTX routines within a DATA Step. I decided to utilize the INTO: statement in the SQL Procedure, because I wanted to exploit the metadata on the four temporary datasets to create macro variables. At runtime, the SAS System creates and maintains metadata on datasets, catalogs, indexes, macros, SAS System options, etc., in read-only virtual tables called *dictionary tables*. One can also access such metadata using the DATA Step and the libref SASHELP (e.g. SASHELP.VCOLUMN). However, I decided, through the SQL Procedure, to use DICTIONARY.COLUMNS to create my macro variables.

The variable NAME in DICTIONARY.COLUMNS corresponds to the name of the column (variable) and the variable MEMNAME corresponds to the name of the table (dataset). In the code that follows, I used information within DICTIONARY.COLUMNS for the four datasets to create 88 macro variables (&COL1VAR1 – &COL4VAR22), each variable corresponding to the name of its respective drug flag within a given time period, plus 22 additional macro variables in the last SELECT statement (&DRUGNAME1 – &DRUGNAME22).

```

PROC SQL NOPRINT;

    SELECT NAME INTO :COL1VAR1 - :COL1VAR22
        FROM DICTIONARY.COLUMNS
        WHERE UPCASE(MEMNAME) EQ "COL1_VARS";

    SELECT NAME INTO :COL2VAR1 - :COL2VAR22
        FROM DICTIONARY.COLUMNS
        WHERE UPCASE(MEMNAME) EQ "COL2_VARS";

    SELECT NAME INTO :COL3VAR1 - :COL3VAR22
        FROM DICTIONARY.COLUMNS
        WHERE UPCASE(MEMNAME) EQ "COL3_VARS";

    SELECT NAME INTO :COL4VAR1 - :COL4VAR22
        FROM DICTIONARY.COLUMNS
        WHERE UPCASE(MEMNAME) EQ "COL4_VARS";

    SELECT SCAN(NAME,1,"3") INTO :DRUGNAME1 - :DRUGNAME22
        FROM DICTIONARY.COLUMNS
        WHERE UPCASE(MEMNAME) EQ "COL2_VARS";

QUIT;

```

If you'll notice, the construct of the last SELECT statement differs from the construct of the previous four; I used the SCAN function to isolate the first five characters from the variables found in the COL2\_VARS dataset (DRUGA3, DRUGB3, DRUGC3, etc.), and stored their values into the macro variables &DRUGNAME1 – &DRUGNAME22. The default delimiter for the SCAN function is a space; since the name of the variable is stored as a text value, the use of "3" as a delimiter tells SCAN to capture the first 5 characters of the variable name into a macro variable before the character "3".

Next, the code below comes from a macro, CATEG\_VARS, that was used to compute proportions using the FREQ Procedure, with a fixed denominator of N = 44. Each run of the FREQ procedure creates 88 datasets, one for each flag, containing counts and percents for each. Once the data were output, datasets for each flag were created using the DATA Step with each dataset containing three variables; COL1 (or COL2, COL3, COL4), and two variables (BLOCK and LINE).

```

proc freq data = &dsn. noprint;
    table &invar.
        /sparse
        out = &invar._propn;
run;

data f_&invar. (keep = col&colnum. block line);
    length col&colnum. $20 ;
    set &invar._propn;

    if &invar. eq 1 then do;
        if count > 0 then col&colnum.=trim(put(count,comma6.)) ||' ('|| trim(put(round(percent,.01)
            else if count eq 0 then col&colnum.=" 0 ( 0.00%)" ;
        line=&line.;
        block=line;
        output;
    end;

run;

```

Once the macro was submitted, I wrapped the macro inside a loop.

```
%MACRO PUT_IT_TOGETHER;

    %DO i = 1 %TO 4;
        %DO j = 1 %TO 22;

            /* Invoke CATEG_VARS macro;
            %CATEG_VARS(dsn=MEDICATION, invar=%%%COL&i.VAR&j., line=%%j., colnum=%%i.);

        %END;
    %END;

%MEND PUT_IT_TOGETHER;
```

For this macro to execute as intended, it was necessary to reference macro variables indirectly, by using both the && and &&&& ampersands. The outer loop references the column number, and the inner loop references the variable number within the column. Resolution of macro variables occur from left to right; when two ampersands (&&) are encountered by the macro processor, they are resolved to a single ampersand (&). Using the first flag, DRUGA1YPREV, as an example, within the first outer, first inner loop, the macro variable &&&&COL&i.VAR&j. ultimately resolves to the value of DRUGA1YPREV; in the first outer loop, second inner loop, &&&&COL&i.VAR&j. resolves to DRUGB1YPREV; in the first outer loop, third inner loop, &&&&COL&i.VAR&j. resolves to DRUGC1YPREV; and so on. In all, 88 datasets were produced, one for each flag.

The step that follows again uses indirect references to macro variables, only this time 22 separate datasets are created, one for each drug, with six variables (COL1 – COL4, block and line), merging information, for each time period, by block and line. The end result was a dataset, for each drug, that summarized frequency of use within each of the four time intervals.

```
%MACRO MERGEFILES;

    %DO k = 1 %TO 22;
        DATA &&DRUGNAME&k. ;
            LENGTH param_txt $200 ;
            MERGE
                F_&&&DRUGNAME&k..1YPREV
                F_&&&DRUGNAME&k..3
                F_&&&DRUGNAME&k..3TO6
                F_&&&DRUGNAME&k..6TO12
            ;
            BY
                BLOCK
                LINE
            ;

            param_txt = put(line, linef.);
        RUN;
    %END;

%MEND MERGEFILES;
```

Once the 22 datasets were created above, a final dataset was created by concatenating data for all drugs, in alphabetical order, using the DATA Step.

In order to bold text and to force line breaks in the report, I used ODS ESCAPECHAR (^). This is done by submitting the following code

```
ods escapechar="^";
```

Specifying a character tells the Output Delivery System (ODS) to interpret whatever follows the caret (^) as a series of control characters. In addition to bolding text and inserting line breaks, you can also use ODS ESCAPECHAR to add superscripts or subscripts, left- or right- justify text, and underline, strikethrough, or overline text. What is displayed below is a style attribute, where one specifies the style to be used (in this case, to bold text). To end it, just put a null style specification at the end. The ^n inserts a line break in the text, which causes the text appearing after it to go to the next line, as seen in the output displayed on the 1<sup>st</sup> page.

```
^S={font_weight=bold}Drug A^S={}^nBrand Name A
```

Using a modified style template, I created an .rtf document with the drug use summary, using PROC REPORT.

```
define style styles.outcome;
  parent=styles.rtf;
  **controls margins**;
  replace Body from Document /
    bottommargin = 0.25in
    topmargin    = 0.25in
    rightmargin  = 0.5in
    leftmargin   = 0.5in;

  **remove color from table, remove gridlines, add underline under column header**;
  style table /
    background = _UNDEF_
    rules      = groups
    frame      = above ;

  **remove color from column header**;
  style header /
    background=_UNDEF_
    cellpadding=3pt;

  **remove color from row header**;
  style rowheader /
    background=_UNDEF_;

  **allow leading blanks in footers**;
  style systemfooter /
    asis=on;

  **change font to not bold, not italicized**;
  replace fonts /
    'titlefont2' = ("Courier New,Courier", 8pt)
    'titlefont'  = ("Courier New,Courier", 8pt)
    'StrongFont' = ("Courier New,Courier", 8pt)
    'EmphasisFont' = ("Courier New,Courier", 8pt)
    'FixedemphasisFont' = ("Courier New,Courier", 8pt)
    'FixedStrongFont' = ("Courier New,Courier", 8pt)
    'FixedHeadingFont' = ("Courier New,Courier", 8pt)
    'BatchFixedFont' = ("SAS Monospace,Courier New,Courier",8pt)
    'FixedFont' = ("Courier New, Courier", 8pt)
    'headingEmphasisFont' = ("Courier New,Courier",8pt)
    'headingFont' = ("Courier New,Courier",8pt)
    'docFont' = ("Courier New,Courier",8pt);
```

Using the .rtf style as a template, I replaced the font, changing it from Times Roman to Courier New, unbolded and de-italicized the text, changed the font size, removed background color from both the column and row headers, removed table gridlines, and adjusted the margin sizes, so that the final version of the table would match what was specified in the table shell.

## CONCLUSION

By taking full advantage of the suite of options available through the SAS<sup>®</sup> System, what appeared at first to be a challenging task was made relatively easy.

## REFERENCES

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Lafler, Kirk Paul 2004. *PROC SQL: Beyond the Basics Using SAS*, Cary, NC: SAS Institute Inc.

Burlew, Michele M. 2002. *SAS<sup>®</sup> Macro Programming Made Easy*. Cary, NC: SAS Institute Inc.